

The binomial theorem

Pascal’s triangle, the general term, and expanding a bracket to any power without multiplying it out.

LESSON 63–64

2 × 40 MIN

ALGEBRA 11, §4.2

P1 · 6.1–6.4

LESSON CLOCK

14'

22'

24'

20'

20'

10'

Pascal's triangle	14 min
The theorem	22 min
The general term	24 min
Approximation	20 min
Practice	20 min
Homework	10 min

LEARNING OBJECTIVES

- Generate Pascal's triangle and relate it to combinations.
- Expand $(a + b)^n$ for a positive integer n .
- Find a specified term of an expansion without expanding it all.
- Use the expansion to approximate a power.

TEXTBOOK

National	pp. 291–302
Cambridge	Pure Mathematics 1, pp. 111–121
Workbook	P1 Exercise 6A–6C

01

Explanation

Pascal’s triangle

Multiply out $(a + b)^n$ for small n and the coefficients form a pattern:

N	COEFFICIENTS	EXPANSION
0	1	1
1	1 1	$a + b$
2	1 2 1	$a^2 + 2ab + b^2$
3	1 3 3 1	$a^3 + 3a^2b + 3ab^2 + b^3$
4	1 4 6 4 1	$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$
5	1 5 10 10 5 1	...

EACH ENTRY IS THE SUM OF THE TWO ABOVE IT

Which is exactly $C(n,k) = C(n-1,k-1) + C(n-1,k)$ from the last lesson. The triangle is the table of combinations, and the coefficients are $C(n, k)$.

The theorem

$$(a + b)^n = \sum C(n,k) a^{n-k} b^k$$

Reading it term by term: the powers of a descend from n to 0 , the powers of b ascend from 0 to n , every term has total degree n , and there are $n + 1$ terms.

$$(2x + 3)^3 = 8x^3 + 3(4x^2)(3) + 3(2x)(9) + 27 = 8x^3 + 36x^2 + 54x + 27$$

THE WHOLE TERM IS RAISED TO THE POWER

In $(2x + 3)^3$ the first term is $(2x)^3 = 8x^3$, not $2x^3$. Bracket the whole of a and the whole of b before applying the coefficients.

The general term

$$\text{the term in } b^k \text{ is } C(n,k) a^{n-k} b^k$$

This is what makes the theorem useful: a single term can be found without expanding the rest.

Example. Find the term in x^5 in $(2x - 3)^8$.

Here $a = 2x$, $b = -3$, $n = 8$. The power of x is $8 - k$, so $k = 3$:

$$C(8,3)(2x)^5(-3)^3 = 56 \times 32x^5 \times (-27) = -48\,384 x^5$$

The term independent of x is found the same way: write the general term, collect the powers of x , and set the total power to zero.

WATCH THE SIGNS

With $(a - b)^n$ take b to be the whole of $-b$. Odd values of k then give negative terms, and the expansion alternates.

Approximation

When x is small, the first few terms of $(1 + x)^n$ approximate it well:

$$(1 + x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$$

Example. $1.02^8 = (1 + 0.02)^8 \approx 1 + 8(0.02) + 28(0.0004) = 1 + 0.16 + 0.0112 = 1.1712$. The true value is 1.17166.

This is the linear approximation of Quarter II with one more term — and each further term improves the accuracy by roughly a factor of x .

02 Worked examples

EXAMPLE 1 Expand $(x + 2)^4$.

① Coefficients 1, 4, 6, 4, 1.

② $x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16$

③ $= x^4 + 8x^3 + 24x^2 + 32x + 16$

ANSWER

$$x^4 + 8x^3 + 24x^2 + 32x + 16$$

EXAMPLE 2 Find the term in x^5 in $(2x - 3)^8$.

$$① \quad k = 3 \text{ gives } x^{8-3} = x^5.$$

$$② \quad C(8,3) = 56$$

$$③ \quad 56 (2x)^5 (-3)^3$$

$$④ \quad = 56 \times 32 \times (-27) x^5 = -48\,384x^5$$

ANSWER

$$-48\,384x^5$$

EXAMPLE 3 Find the term independent of x in $(x^2 + \frac{1}{x})^6$.

$$① \quad \text{General term } C(6,k)(x^2)^{6-k}(x^{-1})^k.$$

$$② \quad \text{Power of } x: 12 - 2k - k = 12 - 3k.$$

$$③ \quad 12 - 3k = 0 \Rightarrow k = 4$$

$$④ \quad C(6,4) = 15$$

ANSWER

$$15$$

03 Interactive model

Build Pascal's triangle to row 6 on the board and leave it there.

Finding one term

The coefficients for $n = 4$ are:

1 4 4 1

1 4 6 4 1

1 3 3 1

1 5 10 10 5 1

$(x + 2)^4$ has constant term:

2

8

16

4

How many terms has $(a + b)^7$?

7

8

14

128

In $(2x + 3)^3$ the first term is:

The term in x^5 of $(2x - 3)^8$ uses:

1.02^8 to three terms:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Binomial	Ikkihad	Двучлен
Binomial theorem	Nyuton binomi	Бином Ньютона
Pascal's triangle	Paskal uchburchagi	Треугольник Паскаля
Binomial coefficient	Binomial koeffitsient	Биномиальный коэффициент
General term	Umumiy had	Общий член
Expansion	Yoyilma	Разложение
Term independent of x	x ga bog'liq bo'lmagan had	Член, не содержащий x
Ascending powers	Darajalar o'sishi bo'yicha	По возрастающим степеням
Approximation	Taqribiy hisoblash	Приближение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The binomial coefficients are:

$$n!$$

$$C(n,k)$$

$$A(n,k)$$

$$n^k$$

$(a + b)^n$ has how many terms?

$$n$$

$$n + 1$$

$$2n$$

$$2^n$$

Every term has total degree:

$$k$$

$$n - k$$

$$n$$

$$2n$$

The sum of the coefficients in $(a+b)^n$ is:

$$n$$

$$n!$$

$$2^n$$

$$n^2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Row 4 of Pascal's triangle

ANSWER $1\ 4\ 6\ 4\ 1$

2. Row 5

ANSWER $1\ 5\ 10\ 10\ 5\ 1$

3. Expand $(x + 1)^3$

ANSWER $x^3 + 3x^2 + 3x + 1$

4. Expand $(x - 1)^3$

ANSWER $x^3 - 3x^2 + 3x - 1$

5. Terms in $(a + b)^6$

ANSWER 7

6. Constant term of $(x + 2)^4$

ANSWER 16

7. Sum of the coefficients of $(a+b)^5$

ANSWER 32

Medium · the standard question

7 PROBLEMS

1. Expand $(x + 2)^4$

ANSWER $x^4 + 8x^3 + 24x^2 + 32x + 16$

2. Expand $(2x - 1)^3$

ANSWER $8x^3 - 12x^2 + 6x - 1$

3. Term in x^3 of $(x + 3)^5$

ANSWER $270x^3$

4. Term in x^2 of $(2x - 1)^6$

ANSWER $60x^2$

5. Coefficient of x^4 in $(1 + x)^9$

ANSWER 126

6. 1.02^8 to three terms

ANSWER 1.1712

7. 0.98^6 to three terms

ANSWER ≈ 0.8858

Hard · stretch and reason

7 PROBLEMS

1. Term in x^5 of $(2x - 3)^8$

ANSWER $-48\,384x^5$

2. Term independent of x in $(x^2 + \frac{1}{x})^6$

ANSWER 15

3. Term independent of x in $(2x - \frac{1}{x^2})^9$

ANSWER -672

4. Coefficient of x^2 in $(1 + 2x)^5(1 - x)^3$

ANSWER 7

5. Find n if the coefficient of x^2 in $(1 + x)^n$ is 45

ANSWER $n = 10$

6. Show the sum of the coefficients of $(2a - b)^n$ is 1

ANSWER Put $a = b = 1$

7. Estimate 1.005^{10} to four decimal places

ANSWER ≈ 1.0511

07 Homework

Homework — 6 tasks

Write the general term before hunting for a specific one.

- Write rows 0 to 6 of Pascal's triangle.
- Expand $(x + 3)^4$ and $(2x - 1)^5$.
- Find the term in x^4 in $(3x + 2)^7$.
- Find the term independent of x in $(x^3 + \frac{2}{x})^8$.

5. Use the first three terms to estimate 1.03^7 and compare with a calculator.
6. Explain in three sentences why the coefficients of $(a+b)^n$ are the combinations $C(n,k)$.